Superstructure & Frame: System Design Description

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Revision	D
Status	APPROVED
Classification	INTERNAL
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1. System overview

The superstructure is the station's load-bearing frame: structural ribs, hull plating sections, the equatorial service ring and the antenna masts.

The system is led by Structural Engineering. It is delivered through 6 work packages with a combined budget of 68,549.52 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
SF-HPS	Hull plating section	SHV-SF-SPEC-0101	SHV-SF-DWG-0201	SHV-SF-BOM-0301
SF-RIB	Structural rib	SHV-SF-SPEC-0102	SHV-SF-DWG-0202	SHV-SF-BOM-0302
SF-ESR	Equatorial service ring	SHV-SF-SPEC-0103	SHV-SF-DWG-0203	SHV-SF-BOM-0303
SF-ANM	Antenna mast	SHV-SF-SPEC-0104	SHV-SF-DWG-0204	SHV-SF-BOM-0304

3. Architecture

3.1 Hull plating section

The hull plating section is specified in SHV-SF-SPEC-0101. Its governing parameter is the plate thickness, set at 6.3 m. The responsible engineer is Jarek Keller.

3.2 Structural rib

The structural rib is specified in SHV-SF-SPEC-0102. Its governing parameter is the rib count, set at 612 ea. The responsible engineer is Ulla Hawthorne.

3.3 Equatorial service ring

The equatorial service ring is specified in SHV-SF-SPEC-0103. Its governing parameter is the ring width, set at 790 m. The responsible engineer is Cedric Keller.

3.4 Antenna mast

The antenna mast is specified in SHV-SF-SPEC-0104. Its governing parameter is the mast height, set at 531 m. The responsible engineer is Edda Parrish.

4. Interfaces

Document	Title
SHV-ICD-SF-HB	Interface Control Document: Superstructure & Frame / Docking & Hangar Bays
SHV-ICD-SF-SY	Interface Control Document: Superstructure & Frame / Shipyard Slipways

Document	Title
SHV-ICD-SF-TR	Interface Control Document: Superstructure & Frame / Internal Transit

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-SF-01	Superstructure & Frame: Design & analysis	Corra Marrek	IN_PROGRESS	98.2%
WP-SHV-SF-02	Superstructure & Frame: Fabricate & install hull plating section	Carys Dravis	IN_PROGRESS	96.1%
WP-SHV-SF-03	Superstructure & Frame: Fabricate & install structural rib	Vela Wade	IN_PROGRESS	88.4%
WP-SHV-SF-04	Superstructure & Frame: Fabricate & install equatorial service ring	Casimir Nevins	IN_PROGRESS	82.4%
WP-SHV-SF-05	Superstructure & Frame: Fabricate & install antenna mast	Ferris Sable	IN_PROGRESS	77.8%
WP-SHV-SF-06	Superstructure & Frame: Integration & test	Orla Kessel	IN_PROGRESS	78.7%

6. Risks

- Supply of hull plating section assemblies depends on a single supplier.
- Supply of structural rib assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.